

A gentle introduction to the file system and the user interface

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Why to bother about Command Line Interfaces?

- **Cleaner approach**: Command line skills help with building repeatable data processes
- **Repeatability**: They facilitate repeatable data manipulation
- **Safer**: Performing repetitive tasks in a CLI makes your work less error prone
- **Precision**: Many bioinformatics tools can be used only by means of a CLI or their CLI version give you access to many more parameters.
- **HPC/Cloud**: Computationally intense work can be only performed on remote or cloud resources, and the preferred access to them is the CLI

CLI and File System

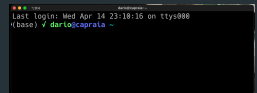
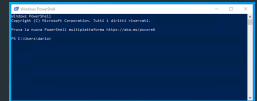
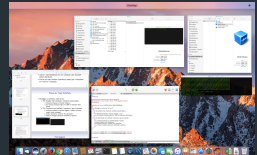
Most of the time spent on the CLI is used to run programs and manage files.

In this mini course you will learn how to use the CLI and how to use it to move in the file system

User Interface: operating system access

Two paradigms: **GUI** vs **CLI**

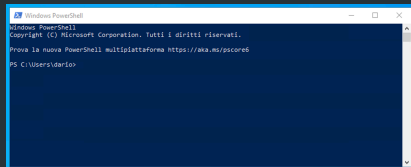
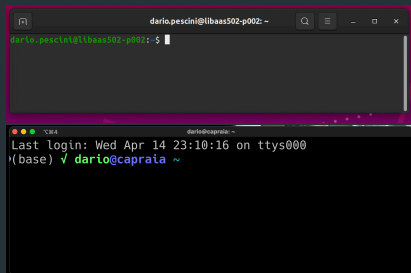
- **GUI** (Graphic User Interface).
 - the interaction is based on windows, buttons, ...
 - it is the most recent (created in the '80s in the Xerox PARC labs)
- **CLI** (Command Line Interface) or Shell.
 - the interaction is based on commands, typed on a keyboard
 - the output: on a screen, in a file, on a printer, ...
 - Useful to execute complex and repetitive actions
 - Philosophy: each specific command do a specific job, combine them into scripts to solve complex problems



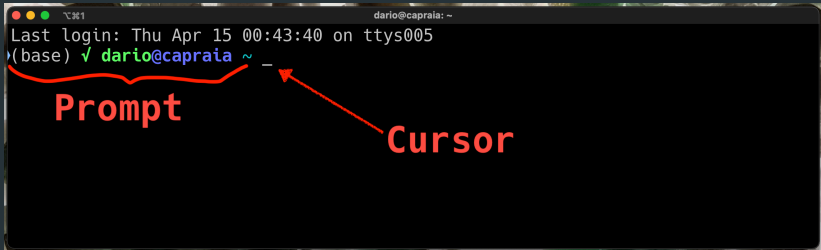
Modern OSs: a shell inside a window

Different OSs, different Shells:

- Linux terminal:
(Bash, zsh, ...)
- OS X terminal/iterm:
(zsh, Bash, ...)
- Windows
PowerShell/cmd.exe



How to interact with the shell



A terminal window titled 'dario@capraia: ~' showing the output of a login. The text 'Last login: Thu Apr 15 00:43:40 on ttys005' is displayed. Below it, the prompt '(base) ✓ dario@capraia ~' is shown, followed by a cursor character '-'. A red bracket underlines the prompt text, and a red arrow points to the cursor character. The word 'Prompt' is written in red below the bracket, and the word 'Cursor' is written in red below the arrow.

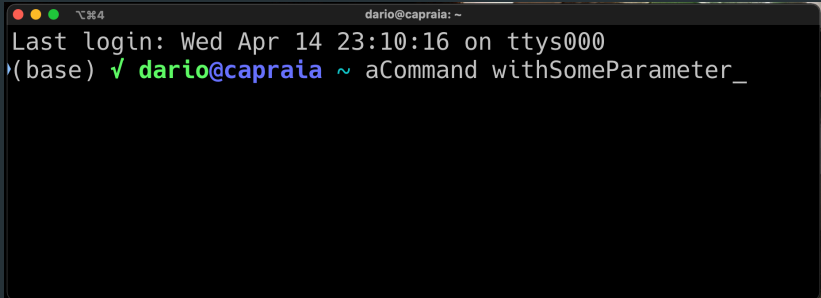
```
Last login: Thu Apr 15 00:43:40 on ttys005
(base) ✓ dario@capraia ~ -
```

Prompt

Cursor

- the shell **prompt** waits you to insert a command at the **cursor**

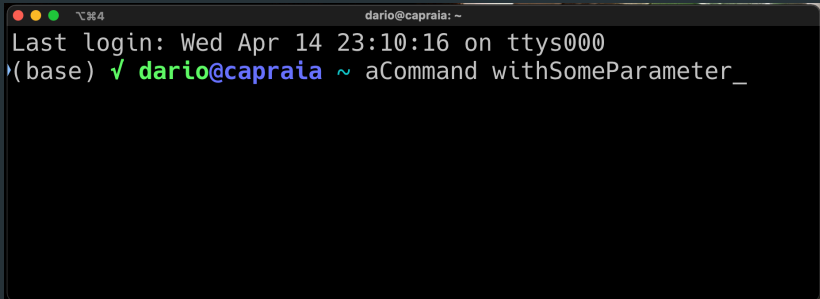
How to interact with the shell



```
darío@capraia: ~  
Last login: Wed Apr 14 23:10:16 on ttys000  
(base) ✓ darío@capraia ~ aCommand withSomeParameter_
```

- a command has this pattern:
`command [ parameter_1 ] [parameter_2 ]... [parameter_N ]`
- where everything between `[` and `]` is optional

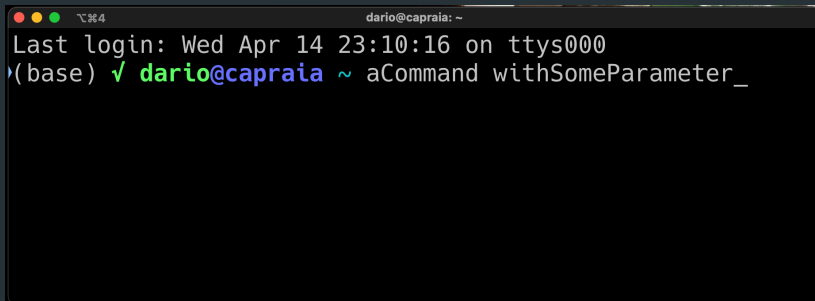
How to interact with the shell



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- a command has this pattern:
`command [ parameter_1 ] [parameter_2 ]... [parameter_N ]`
- where everything between `[` and `]` is optional
- manual: `man command` open the documentation for the command

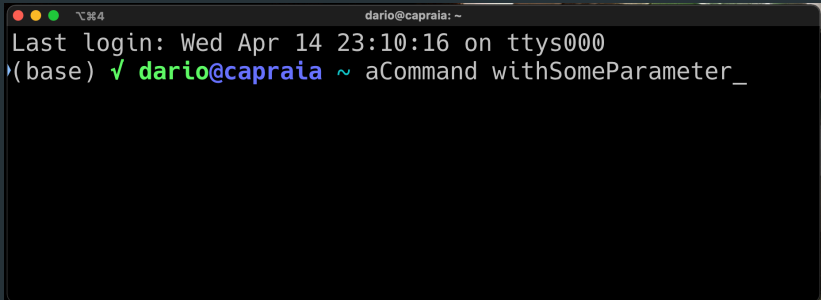
How to interact with the shell



```
darío@capraia: ~  
Last login: Wed Apr 14 23:10:16 on ttys000  
(base) ✓ darío@capraia ~ aCommand withSomeParameter_
```

- tab completion: start to digit a command and then press → (tab)

How to interact with the shell



A terminal window titled 'dario@capraia: ~' with a dark background. The window shows the following text: 'Last login: Wed Apr 14 23:10:16 on ttys000' followed by a prompt '(base) ✓ dario@capraia ~ aCommand withSomeParameter_'. The prompt is on a new line. The window has standard macOS window controls (red, yellow, green buttons) in the top left corner.

- tab completion: start to digit a command and then press → (tab)
- history: it is possible to navigate the already used commands by means of ↑ and ↓

The File System

The **File System** is the Operating System (OS) component dedicated to the **permanent storage** of the information.

- It controls **how** the information is stored and accessed in storage disks
- It exposes to the user simple interfaces to **manage** (write, delete, copy, ...) the stored information

File system

From a user point of view, it is built upon two building blocks:

- **file** : uniform logic representation of correlated information
 - atomic entity
 - collection of correlated information
- **directory** : collection of information used to organize and access files meta-data
 - collection of files
 - access point to files

File Attributes - meta data

- **name**: A string used to identify a specific file within the file system
- **location**: the position of the file on the storage unit
- **kind**: It is used by some OSs to asses the kind of the file content
- **dimension**
- **date-time**: information relatives to time of the creation or last modification
- **ownership**: owner, group. Used to manage authorizations and accounting
- **protection**: file access policies

File, Directory and Meta dati

When you list the content of a folder (i.e. directory) the shell exposes you files and folders meta data:

```
(base) ✓ dario@capraia dario ls -lh
total 872
drwxr-xr-x  2 dario  staff    64B Apr 15 22:25 A01
drwxr-xr-x  2 dario  staff    64B Apr 15 22:25 A02
-rw-rw-r--  1 dario  staff  416K Apr 15 22:25 CFCodiceComuniItalia.csv
-rw-rw-r--  1 dario  staff  233B Apr 15 22:25 CFTabellaDispari.tsv
-rw-rw-r--  1 dario  staff  228B Apr 15 22:25 CFTabellaPari.tsv
-rw-rw-r--  1 dario  staff  119B Apr 15 22:25 CFTabellaResto.tsv
-rw-----  1 dario  staff   4.0K Apr 15 22:25 CFcalcolo.py
```

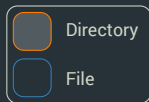
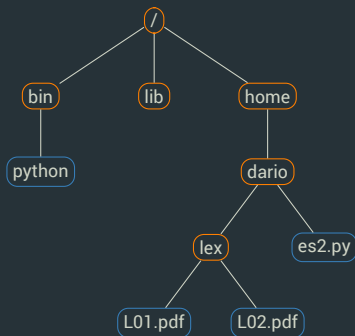
dir	Owner Perm.	Group Perm.	Others Perm.	Owner Name	Group Name	Size	Last Modified	Name
d	rwX	r-X	r-X	dario	staff	64B	Apr 15 22:25	A01

Permissions:

- **d** directory
- **r** read
- **w** write
- **X** execute/traverse

File System

In the file system, **files** and **folders** are structured as a tree:



File System

Also from the user perspective the "file system" is structured as a tree:

Favorites	Name	Date Modified	Size	Kind
↳ dario	3StepLinActivation	25 May 2015, 17:17	13 KB	Folder
↳ Desktop	coagulation	10 Mar 2014, 12:52	2,4 GB	Folder
↳ Works	EColi	6 Sep 2016, 11:43	6,1 MB	Folder
↳ Downloads	JCP	28 Aug 2016, 15:25	204,03 GB	Folder
↳ Lezioni	mappaTenda	9 Mar 2015, 11:56	23,2 MB	Folder
↳ ownCloud	MatlabTesiTumolo	18 Jun 2015, 13:04	32,5 MB	Folder
↳ Dropbox	NaCoDynCIB8	2 Jun 2016, 11:30	622,8 MB	Folder
↳ Google Drive	↳ Filt	11 Jun 2016, 15:30	338 MB	Folder
↳ Simulation	allFluxesDendro.pdf	11 Jun 2016, 17:22	14 KB	PDF
↳ workspace	cbFluxesDendro.pdf	11 Jun 2016, 17:16	14 KB	PDF
↳ labview	img	7 Jun 2016, 14:09	618 KB	Folder
↳ matlab	NacoGlcExpsF01.h5	2 Jun 2016, 11:25	28,2 MB	HDF5 Data File
↳ DLP	NacoGlcExpsF001.h5	2 Jun 2016, 12:04	28,1 MB	HDF5 Data File
↳ Documents	NacoGlcExpsF001#1e4.h5	2 Jun 2016, 17:45	280,9 MB	HDF5 Data File
↳ Applications	NACOirr.xml	2 Jun 2016, 11:08	46 KB	XML Document
↳ Testi	scatterCBRespFerm.pdf	11 Jun 2016, 17:22	125 KB	PDF
↳ Pictures	NACOirr.xml	2 Jun 2016, 11:08	46 KB	XML Document
↳ Movies	noFilt	3 Jun 2016, 11:27	284,7 MB	Folder
↳ Music	img	2 Jun 2016, 22:59	2,6 MB	Folder
↳ AirDrop	NacoGlcExps#1e4.h5	2 Jun 2016, 13:23	282 MB	HDF5 Data File
↳ All My Files	NACOirr.xml	2 Jun 2016, 11:03	46 KB	XML Document
↳ iCloud Drive	tmp	2 Jun 2016, 14:10	212 bytes	Text Document
↳ Deleted Users	ODH	25 May 2015, 17:18	20 KB	Folder
Devices	↳ input	2 Apr 2015, 09:08	2 KB	Folder
↳ vulcano	ODH.dot	2 Apr 2015, 09:10	4 KB	Microsoft Word Document
↳ Macintosh HD	ODH.prj	2 Apr 2015, 09:08	137 bytes	Document
↳ Remote Disc	ODH.xml	2 Apr 2015, 09:09	9 KB	XML Document
↳ HD-PATU3	psoGlbbs	25 Nov 2015, 08:42	9,2 MB	Folder
	PulseGen	26 Jul 2014, 11:38	783,7 MB	Folder
	RAS	7 Nov 2014, 12:15	1,03 GB	Folder
	RNG_MAP	20 May 2015, 12:08	167 KB	Folder
	RNG_MAPold	10 Mar 2015, 12:22	61,1 MB	Folder
	SAS2R	23 Mar 2015, 18:03	14 KB	Folder
	schlogi	4 Nov 2016, 11:41	164 bytes	Folder
	SecondRound	8 Oct 2014, 09:20	551,8 MB	Folder
	testOliver30	7 Feb 2015, 08:54	369 bytes	Folder
	triSome	26 Jul 2014, 11:38	48,41 GB	Folder

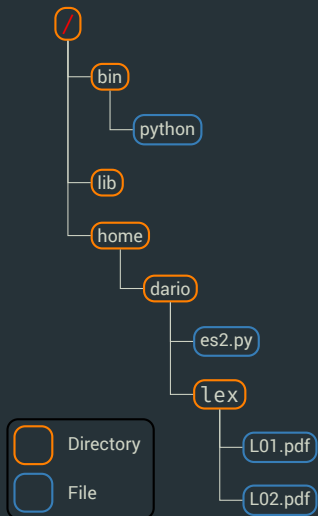
File system: jargon

Given the file `L01.pdf` in the file system on the right,



- `L01.pdf` is the file **name** (or base name)
- `pdf` is the file **extension**
- `/home/dario/lex/` is the **directory** that contains the file
- `/home/dario/lex/L01.pdf` is the **absolut path** (*fully qualified name or full path*)
- **relative path**
 - given that `/home/dario/` is the current **working directory**
 - `lex/L01.pdf` is the **relative path** of `L01.pdf` w.r.t. the working directory

File system: jargon



- It is possible to reason about folders name and path, in an analogous way to what we have done for files.
- Given the path `/home/dario/lex/L01.pdf`
 - `/` is the **root**, folder: the origin of every and each path in the file system
 - the other `/` are file and folders visual separators
 - the symbol `/` is called *slash*

File Systems in Unix and Windows

These two families of OSs use different folder separator / slash versus \ back slash, this can hamper your work.

OS	Root Directory	Directory Separator	Examples
Unix	/	/	/home/user/docs/Letter.txt
Win	[drive letter:]\	\	C:\User\MrX\Letter.txt \User\MrX\Letter.txt MrX\Letter.txt

- The [] wrapping the *drive letter*: means that you can omit the Drive letter indication

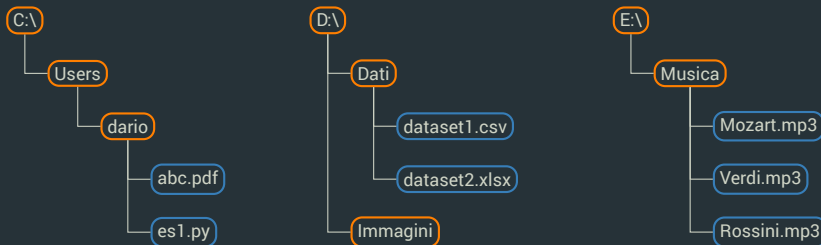
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Unix	/	/	/home/user/docs/Letter.txt
Win	[drive letter:]\	\	C:\User\MrX\Letter.txt \User\MrX\Letter.txt MrX\Letter.txt

- The [] wrapping the *drive letter*: means that you can omit the Drive letter indication
- If in windows you use the PowerShell you can ignore this difference.

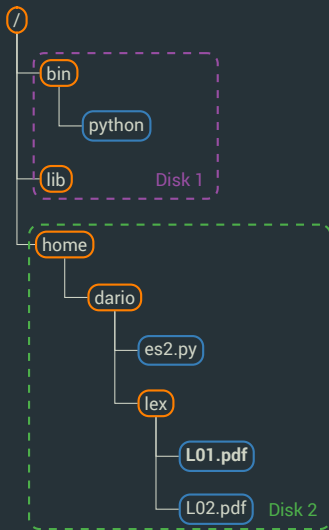
Windows: Drives and Forests



- The windows file system is a forest of trees
- Each drive is the root of a distinct tree.
- To change the tree you have to call it: E :

Unix: Drives on a Tree

- In the Unix file systems, apparently there is no trace of the different drives.
- The devices are directly *mounted* on the folders tree.
- This transparent approach make it easier to move folders to different drives.



How to navigate in your file system

- `pwd` - where am I ?
- `ls` - what is in here ?
- `cd` - take me there !

Paths recap

- `/` root
- `.` here
- `~` home
- `..` level up
- `-` where I was before